Requirements Engineering Principles & Practices

RCS 305 ? Software Engineering | Riara University Revision Resource

1. Foundational Objectives

Requirements Engineering (RE) is the systematic discipline of discovering, documenting, analyzing, and maintaining software requirements throughout system evolution. Poor requirements analysis accounts for over 60% of critical project failures.

2. The Requirements Engineering Process

The RE lifecycle comprises four intertwined activities:
? Inception & Elicitation: Engaging stakeholders through structured interviews, workshops, ethnography, and domain analysis.
? Analysis & Negotiation: Resolving conflicting stakeholder goals, assessing technical feasibility, and establishing boundary scopes.
? Specification: Drafting structured specifications adhering to standard schemas.
? Validation: Reviewing documents with users to guarantee correctness, consistency, and completeness.

3. Functional vs. Non-Functional Requirements

? Functional Requirements (FR): Explicitly define system behavior, calculations, data transformations, and business rules (e.g., 'The platform shall authenticate students via JWT bearer tokens').
? Non-Functional Requirements (NFR): Constrain operational qualities, including latency, throughput, availability (99.9% uptime), data encryption standards (AES-256), and WCAG 2.1 accessibility.

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